Weihan Li

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Education

Georgia Institute of Technology, PhD in Machine Learning — Sept 2023 – Present

Georgia Institute of Technology, Master of Science in Computer Science — 2023 – 2025

- **Ph.D. Advisor**: Prof. Anqi Wu

Zhejiang University, Research Assistant — Sept 2020 – May 2023

- **Advisor**: Prof. Gang Pan

Zhejiang University, Bachelor in Computer Science and Technology — Sept 2016 – June 2020

Research Summary

My current research is in **robot learning for dexterous bimanual manipulation**: how multi-fingered robot hands can learn diverse manipulation skills from human demonstrations, and how skills learned in simulation can transfer reliably to real hardware. Previously, I studied how multiple brain regions communicate and how animal behavior can be understood at scale, with applications to brain disorders and brain-computer interfaces.

Research Experience: Robot Learning

Dexterous Bimanual Manipulation from Human Demonstrations — Aug 2026 – Present

Platform: bimanual 20-DoF dexterous hands (Wuji, Sharpa) on a Vega humanoid

- Turning human hand–object demonstrations into physically feasible robot demonstrations through kinematic and physics-based retargeting.
- Learning residual reinforcement-learning policies on top of the retargeted demonstrations to perform contact-rich manipulation robustly.
- Working toward sim-to-real transfer by aligning simulation with real-robot physics and evaluating policies against hardware-readiness criteria.

Research Experience: AI for Neuroscience and Behavior

Mouse Behavior Analysis — Sep 2025 – Jul 2026

- **ResiFuse** (NeurIPS 2026; first author): Pose–video fusion that adds video features only on frames with large quantization residuals in a pose-based motion encoder; outperforms pose- and video-based baselines on most tasks across three mouse datasets, with the largest gains in the most socially complex setting.
- **MICE** (NeurIPS 2026 Spotlight): A hierarchical self-supervised foundation model for mouse behavior, pretrained jointly on six mouse-pose datasets; a Perceiver-style social encoder lets a single checkpoint handle any number of animals.
- **QDMouse4M** (NeurIPS 2026 E&D Track): A six-view dataset of over four million frames of freely moving mice, with 2D and 3D keypoint labels from subdermal quantum-dot fluorescence markers, for benchmarking pose estimation and behavior analysis.
- **BehaviorVLM** (preprint; co-first author): A finetuning-free framework for pose estimation and behavioral understanding that guides pretrained vision–language models through explicit, verifiable reasoning steps.

Multi-Region Brain Communication — Sep 2023 – May 2026

- **MBRILA** (open-source software; creator): A general Gaussian-process state-space (GP-SSM) framework in which any Gaussian-process model can be converted into an equivalent state-space form with scalable inference; MRM-GP and ADM are built on it. MIT license, `github.com/BRAINML-GT/MBRILA`.
- **ADM** (ICML 2025 **Oral**; first author): An adaptive delay model that combines Gaussian processes with state-space models to learn time-varying, directional communication across brain regions, with parallel-scan inference that reduces the time complexity from $O(T^3)$ to $O(\log T)$.

- **MRM-GP** (ICML 2024; first author): A linear dynamical system that mirrors a multi-output Gaussian process with explicit frequencies and phase delays, recovering directional, frequency-specific communication across brain regions in linear time.

Statistical Models of Neural Activity Jan 2022 – 2026

- **FacRNN** (ICML 2026 Spotlight): A generative low-rank RNN framework that encourages group-wise independence among latent dynamics, yielding more interpretable neural latent trajectories and low-rank connectivity.
- **MIG-Vis** (ICLR 2026): Mutual-information-guided diffusion that visualizes the visual-semantic features encoded by disentangled neural latent groups in macaque IT cortex.
- **BeNeDiff** (NeurIPS 2024): Behavior-relevant, disentangled neural subspaces, with video diffusion models that generate behavior videos revealing what each neural factor encodes.
- **POGLM** (ICML 2024): A differentiable partially observable GLM with forward–backward message passing for learning neural connectivity in the presence of hidden neurons.
- **VIS** (ICLR 2024 Spotlight): Variational importance sampling that maximizes the log-likelihood of latent variable models using a proposal obtained by minimizing the forward χ^2 divergence.
- **HDPP** (NeurIPS 2022; first author): A hierarchical Dirichlet point process that detects neural sequences online with particle filters and introduces new sequence types on the fly.

Publications

Learning When Visual Context Matters for Mouse Behavior Analysis **NeurIPS 2026** [link]
Weihan Li, Jingyang Ke, Qinheng Pu, Yule Wang, Chengrui Li, Anqi Wu

MICE: Multi-animal Interaction Context Encoder — A Hierarchical Foundation Model for Mouse Behavior **NeurIPS 2026 Spotlight**
Yen-Shuo Su, ***Weihan Li***, Anqi Wu

QDMouse4M: A Multi-View 3D Mouse Spontaneous Behavior Dataset with Quantum-Dot Markers **NeurIPS 2026 E&D Track**
Jingyang Ke, Amartya Pradhan, Xueling Zhang, ***Weihan Li***, Anqi Wu, Jeffrey E. Markowitz

BehaviorVLM: Unified Finetuning-Free Behavioral Understanding with Vision-Language Reasoning preprint [link]
Jingyang Ke[*], ***Weihan Li***[*], Amartya Pradhan, Jeffrey Markowitz, Anqi Wu
[*]*Equal contribution; authors listed alphabetically.*

A Factorized Low-Rank RNN Framework for Uncovering Independent Neural Latent Dynamics and Connectivity **ICML 2026 Spotlight** [link]
Chengrui Li, Yunmiao Wang, Yule Wang, ***Weihan Li***, Dieter Jaeger, Anqi Wu

Uncovering Semantic Selectivity of Latent Groups in Higher Visual Cortex with Mutual Information-Guided Diffusion **ICLR 2026** [link]
Yule Wang, Joseph Yu, Chengrui Li, ***Weihan Li***, Dieter Jaeger, Anqi Wu

Learning Time-Varying Multi-Region Brain Communications via Scalable Markovian Gaussian Processes **ICML 2025 Oral** [link]
Weihan Li, Yule Wang, Chengrui Li, Anqi Wu

Exploring Behavior-Relevant and Disentangled Neural Dynamics with Generative Diffusion Models **NeurIPS 2024** [link]
Yule Wang, Chengrui Li, ***Weihan Li***, Anqi Wu

Multi-Region Markovian Gaussian Process: An Efficient Method to Discover Directional Communications Across Multiple Brain Regions **ICML 2024** [link]
Weihan Li, Chengrui Li, Yule Wang, Anqi Wu

A Differentiable Partially Observable Generalized Linear Model with Forward-Backward Message Passing **ICML 2024** [link]
Chengrui Li, ***Weihan Li***, Yule Wang, Anqi Wu

Forward χ^2 Divergence Based Variational Importance Sampling **ICLR 2024 Spotlight** [link]
Chengrui Li, Yule Wang, ***Weihan Li***, Anqi Wu

Online Neural Sequence Detection with Hierarchical Dirichlet Point Process **NeurIPS 2022** [link]
Weihan Li, Yu Qi, Gang Pan

Efficient Point-Process Modeling of Spiking Neurons for Neuroprosthesis **EMBC 2021** [link]
Weihan Li, Cunle Qian, Yu Qi, Yiwen Wang, Yueming Wang, Gang Pan

Technical Skills

Robotics & Simulation: Isaac Lab, MuJoCo, Motion Retargeting, Sim-to-Real Transfer

Machine Learning: PyTorch, Reinforcement Learning, Vision-Language Models

Languages: Python, C++

Teaching and Academic Services

Teaching Experience: TA for Georgia Tech CSE-8803 Statistical Machine Learning Models for Neural and Behavioral Data Analysis.

Conference Reviewer: NeurIPS 2023/2024/2025/2026, ICML 2024/2025/2026, ICLR 2024/2025/2026.